

**New trading
relationships can
change the
dominant vehicle.**



**Centre for Blockchain
Technologies**

The Making of Dominant Vehicle Currencies

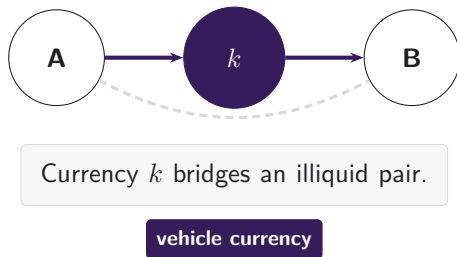
Evidence from DeFi
Jiahua Xu

A cross-border payment needs someone to make the FX market

THE LIQUIDITY-PROVIDER PROBLEM

A payment provider receives one currency but must deliver another it does not hold.

Project Nexus requires an FX provider.
Project Rialto uses a vehicle when direct exchange is unavailable.



Winning the middle concentrates trading and liquidity in the vehicle.

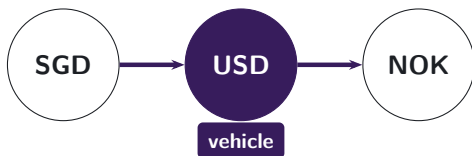
Note: Project Nexus states that every cross-border, cross-currency payment needs an actor willing and able to exchange the currencies. Project Rialto experimentally uses a vehicle currency when direct exchange is unavailable. The paper studies the intermediary asset observed inside DEX pool routes.

Pool routes reveal the vehicle currency

SINGAPORE PAYMENT ANALOGY

A Singapore-dollar payment to Norway may be converted through US dollars.

Conventional turnover data show two legs, not the customer's connected route.



Pool logs reveal the route.

DeFi is on-chain finance executed by smart contracts; decentralised exchanges (DEXs) trade tokens through liquidity pools. **Stablecoins** are tokens designed to track a reference currency, usually the US dollar.

Note: The currency labels illustrate a possible indirect conversion, not an observed customer route. Conventional FX records report turnover in each currency pair but generally do not link the two legs to one instruction. The DEX setting preserves the transaction-level pool route, so the intermediary asset is observed directly.

The route panel links pool-level swaps

475 million

pool-level swaps

2,798

calendar dates

9

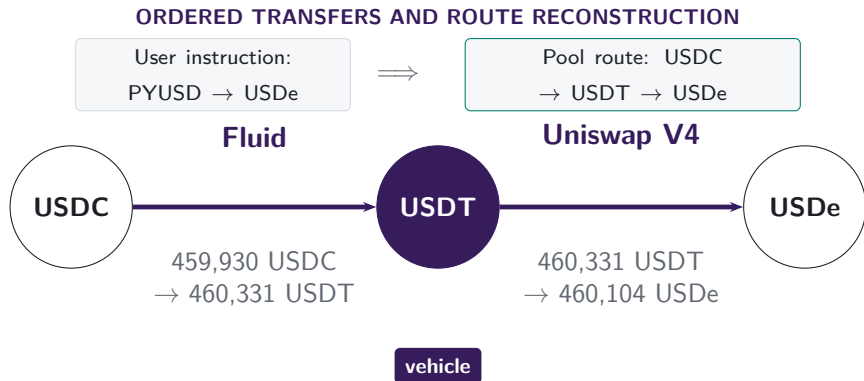
DEX deployments

November 2018–June 2026

Uniswap v1, v2, v3, and v4; SushiSwap v2 and v3; Curve; Balancer; and Fluid.

Note: The exchange families were selected for historical Ethereum volume. The nine deployments span major constant-product, concentrated-liquidity, stable-swap, weighted-pool, singleton, and integrated-liquidity designs. Other Ethereum venues remain outside the panel. The appendix compares the observed sources with total Ethereum DEX volume. The exact exchange registry maps every observed Uniswap v1 exchange to its token.

Inside the pools, USDT links USDC to USDe



The ordered endpoint pair, or pair, is $\text{USDC} \rightarrow \text{USDe}$. Its legs are $\text{USDC} \rightarrow \text{USDT}$ and $\text{USDT} \rightarrow \text{USDe}$; USDT is the vehicle.

Note: The route is the connected pool-swap component of transaction `0xbda1d07f...9bf67ad9` on 10 January 2026. The pool component begins at USDC. It does not reveal whether PYUSD became USDC in an unobserved pool or through the executor's inventory.

The full route and the one-vehicle choice answer different questions

MEASUREMENT

- Every complete route contributes each currency used in an intermediary position.
- Exact two-leg routes isolate one vehicle choice.



All route lengths

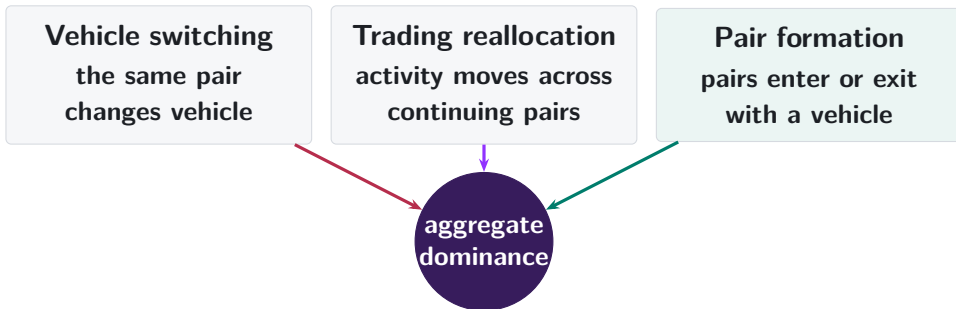
Presence on a route and share of all intermediary positions

One vehicle per route

Exact two-leg routes support decomposition and choice tests

Note: Intermediary-position shares sum to 100%. Route-participation shares can exceed 100% because long routes use several currencies. Exact two-leg routes isolate one vehicle choice; longer routes can use both families.

Aggregate dominance can change in three ways



Note: The decomposition is exact. The next step asks why a pair keeps its first vehicle and when a competing two-leg bridge becomes deep enough to attract routes.

Protocol design changes the route set in steps

UNISWAP ON ETHEREUM



V2 opens vehicle choice; later versions change how liquidity is supplied.

Note: First-swap dates. The next slide isolates V1 to V2: compulsory ETH routing ends while the inherited market structure remains.

Native-asset pairing persists after it becomes optional

V1: PROTOCOL RULE



**217,003 token-to-token trades
pass through ETH**

Every pool is an ETH-token pool, so the architecture imposes the vehicle.

V2: POOL FORMATION

95.5%

of single-leg V2 trades use a WETH pool in 2026

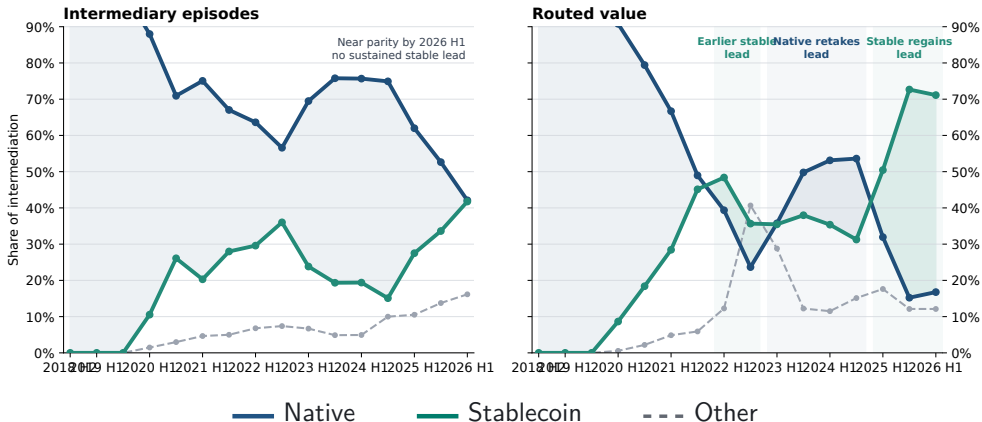
97.9%

of token combinations first traded in 2026 include WETH

Note: V1 links token-to-ETH and ETH-to-token swaps within one transaction. V2 reports legs and newly traded token combinations on that venue. Early V2 inherited liquidity, users, and routing conventions from V1, so part of the initial persistence is mechanical; later pool formation shows how long that structure lasts.

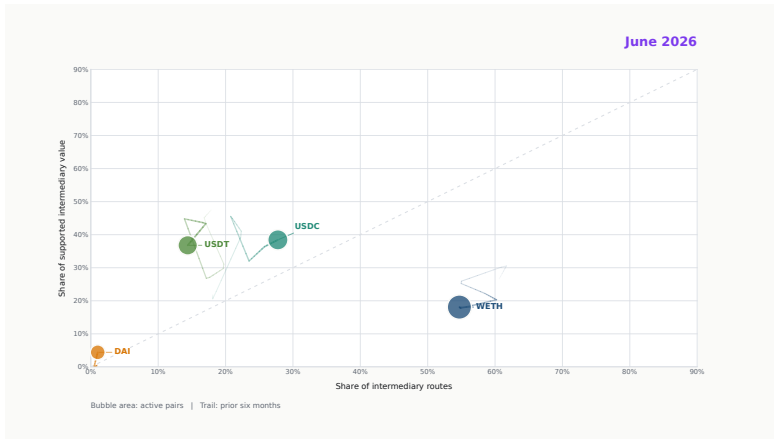
Stablecoins regain the routed-value lead by 2026 H1

ALL ROUTE LENGTHS



Note: Each complete route contributes once for every intermediary position, including routes with more than two legs. Every point covers six calendar months; the final point is 2026 H1. Routed value requires source, intermediary, and destination values to agree within 20%.

Vehicle leadership turns over through time



[Click the film to play in Adobe Acrobat](#)

Note: The film follows WETH, USDC, USDT, and DAI in route participation, supported value, and pair breadth. Each frame retains six months of history, so no single frame contains the full transition. The poster is the PDF fallback.

Pair formation supplies most of the rotation

EXACT DECOMPOSITION IN PERCENTAGE POINTS (PP): 2024 H1 TO 2026 H1

Same continuing
pair

−0.1 pp

Trading reallocation

+8.4 pp

Pairs in one period

+17.8 pp

Stablecoin
share

2024 H1
16.9% → 2026 H1
42.4%

The aggregate rises +25.5 pp; net switching inside continuing pairs is −0.1 pp.

Note: Exact two-leg native-or-stable vehicle routes on matched January–June dates. A further −0.5 pp comes from the changing weight of continuing versus period-specific pairs. Components use unrounded values and sum to the aggregate change.

The first vehicle is sticky

PREDICTIVE PERSISTENCE AFTER PAIR ENTRY

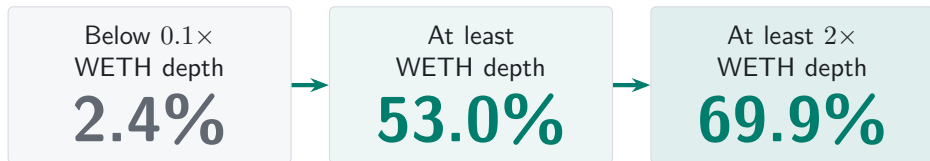


The first vehicle remains a focal point for the new trading relationship.

Note: 599,870 non-WETH pairs observed by route scope. Regressions control for entry size, cohort year, endpoint class, direct-route share, and route complexity. Standard errors are clustered by entry date. "Sticky" means predictive persistence in observed route use; it does not identify trader inertia.

A shallow stablecoin bridge attracts little route flow

FIRST 30 DAYS AFTER PERSISTENT STABLECOIN SUPPORT

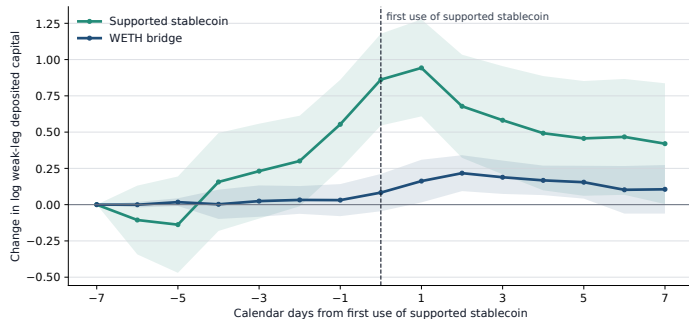


First-month use rises from 42.6% below $0.1\times$ WETH depth to 84.1% above it.

Note: Depth is prior-calendar deposited capital on the weaker leg. The route-share comparison uses 11,327 pair-days across 855 bridge events. Support, depth, and first use refer to the same stablecoin set. Capital and route demand are jointly determined.

Bridge capital builds before first use

SUPPORTED STABLECOIN FIRST USED AT DAY ZERO



BEFORE FIRST USE

+0.86

stablecoin bridge, day -7
to 0

AFTER FIRST USE

-0.44

stablecoin bridge, day 0
to +7

Capital builds into first use, then partly unwinds.

Note: 246 pair events with first use 7 to 119 days after persistent support. Relative to WETH, capital changes +0.78 before and -0.46 after first use. 92.5% of first-use capital lies in pools active one week earlier, so the rise mainly deepens existing bridges. Capital is lagged one day; intervals cluster by pair and adoption date. First use is endogenous.

Price competition usually changes the venue

SHARE OF ROUTES WITH A QUOTE MORE THAN 1 BP BETTER THAN EXECUTION

1. SAME VEHICLE,
USED VENUES



2. SAME VEHICLE,
ALL EXACT VENUES



3. ANY VEHICLE
OR DIRECT

6.6%

of observed routes

44.5%

of observed routes

46.4%

of observed routes

12.0%

entry vehicle retained when the
challenger price-leads

-1.2 pp

change in stablecoin vehicle share

Note: Routes are repriced for the same pair and input immediately before execution. The sample contains 777,203 routes on 73 monthly dates across exact Uniswap v2, SushiSwap v2, and Uniswap v3 states; minimum input is \$100 and the impact limit is 5%. At least 120 days after entry, the vehicle is retained on 99.5% of routes when it price-leads and 12.0% when a challenger leads. Quotes are gross of gas.

New relationships can reorder currency dominance

Dominance concentrates markets

The vehicle links pairs to common liquidity and issuer conditions.

Network growth carries the rotation

Formation and reallocation carry the rotation.

Early use and two-leg depth matter

Entry predicts use; route share rises with depth.

New pairs choose the bridges that shape dominance.

Appendix map

Definitions and data

- A1 Route units and vehicle status
- A2 Asset types and backing regimes
- A3 Sample construction and exclusions

Reconstruction and validation

- A4 Exact transaction state
- A5 Protocol-specific state
- A6 Forced routes in V1

Robustness

- A7 Timing and venue scope
- A8 Count and value weighting
- A9 Observed route endpoints

Market structure

- A10 Capital, inventory, and depth
- A11 Coordination and integration
- A12–A13 References

Further results

- A14 Graph position and realised use
- A15 Endpoint demand
- A16–A17b Pair composition
- A18 Local and network depth
- A19 Pool activation and growth
- A20 V4 participation

Pool data may start after the user instruction

USER INSTRUCTION RECORDED BY THE EXPLORER

Transaction details # 0x

⚡ Swap 460.00K PYUSD for 460.10K USDe on 0x 0x

THE DECISIVE SETTLEMENT TRANSFERS

USDC
ERC-20 **Transfer**

Market Maker: 0x51c7..... → Mainnetsetter

USDC
ERC-20 **Transfer**

Mainnetsetter → Fluid: Liquidity

The maker supplies the USDC used by the Fluid leg. The instruction is PYUSD → USDe.

Note: Transaction 0xbda1d07f...9bf67ad9, 10 January 2026. The data do not reveal whether PYUSD–USDC is another vehicle leg or executor inventory. Disconnected groups comprise 6.2% of reconstructed components in 2026. Treating each internally valid component as a separate route raises stablecoin count dominance from 42.1% to 43.7%.

A1. One reconstructed route is one unit



one route unit
two swap legs

- Each linked $i \rightarrow k \rightarrow o$ route is counted once.
- Asset k is classified as the intermediary on that route.
- Dominance is the share of route units or supported value carried by k .

How we reconstruct a route from one transaction

Direct swap

one transaction



Sequential vehicle route

one transaction



Parallel split

one transaction



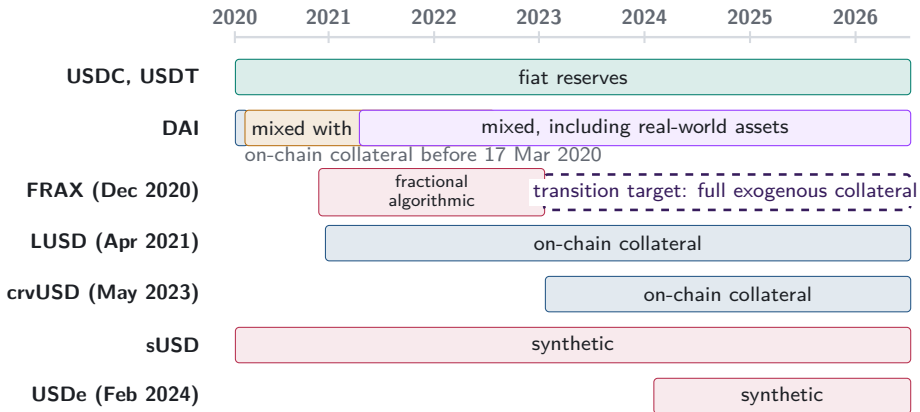
Round trip

one transaction



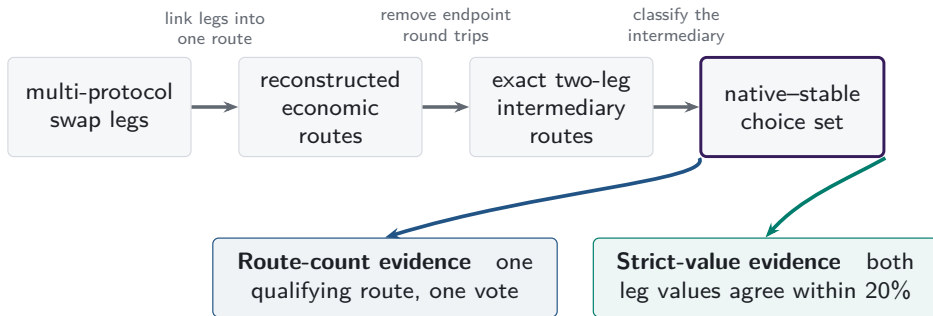
Vehicle choice directs indirect order flow toward particular assets and pools. Transaction-level routes make that choice visible inside aggregate turnover.

A2. Stablecoin backing changes over time



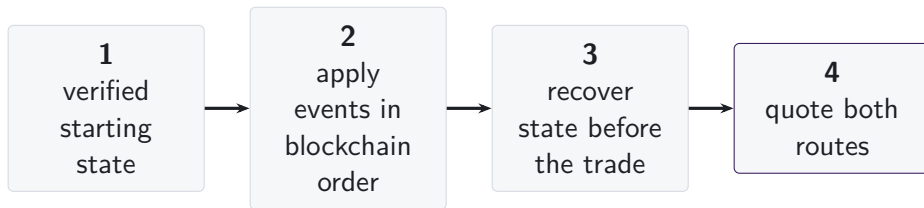
Note: Bars begin at public launch or the sample start for older tokens. DAI's dated regimes follow the Maker Governance actions that added USDC-A and activated RWA001-A. FRAX's dashed arrow follows FIP-188 and denotes a target, not observed collateral.

A3. One route universe supports two measurements



Token combinations inside multi-asset pools can supply route legs. The V1/V2 architecture comparison isolates the venue rule within the shared route panel; dollar support changes value weighting.

A4. State is reconstructed immediately before execution



- Pool events are applied in blockchain order from a verified starting state.
- Each route is quoted using the pool state immediately before the transaction.
- Missing state intervals are excluded, and their share of trading volume is reported.

A5. Each AMM family has a different state vector

	reserves or active liquidity	ticks	amplification	weights or scaling	rates or oracles	fees	hook-dependent pricing
Constant product	■	—	—	—	—	■	—
Concentrated liquidity	■	■	—	—	—	■	—
StableSwap families	■	—	■	—	■	■	—
Weighted pools	■	—	—	■	■	■	—
V4 singleton	■	■	—	—	—	■	□

Executable-quote validation

■ recovered pricing input

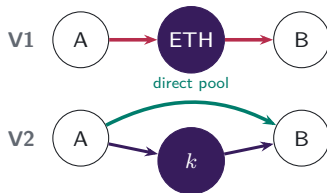
□ hook-dependent pricing unavailable

Protocol mechanisms studied

V4 singleton flash accounting and netting remain in scope; A5.3 shows the settlement mechanism.

A5.1. Pool formation determines available paths

V1 PROTOCOL RULE $\rightarrow$ V2 MARKET CHOICE



V1 admits only ETH-token pools. V2 permits any ERC-20 pool, so market creation determines whether direct and vehicle paths coexist.

Which paths exist?

Economic implication: pool creation changes the feasible route set.

A5.2. Executable depth is path-specific

V3 CONCENTRATED-LIQUIDITY POSITIONS



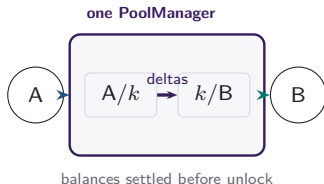
LPs choose a price range and fee tier for each atomic pool.
Positions covering the current price determine active liquidity
in the direct pool and each vehicle spoke.

Which path offers depth at this trade size?

Economic implication: executable depth is path- and size-specific.

A5.3. Shared accounting moves route settlement

V4 SINGLETON AND LOCK ACCOUNTING



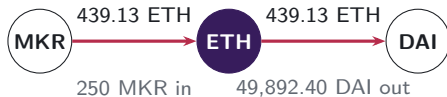
One PoolManager holds all pools. During a lock, the two route legs update balance deltas; the caller settles its obligations before the lock ends.

Where is the route settled?

Economic implication: shared accounting changes where route obligations settle.

A6. V1 forced routes have an on-chain signature

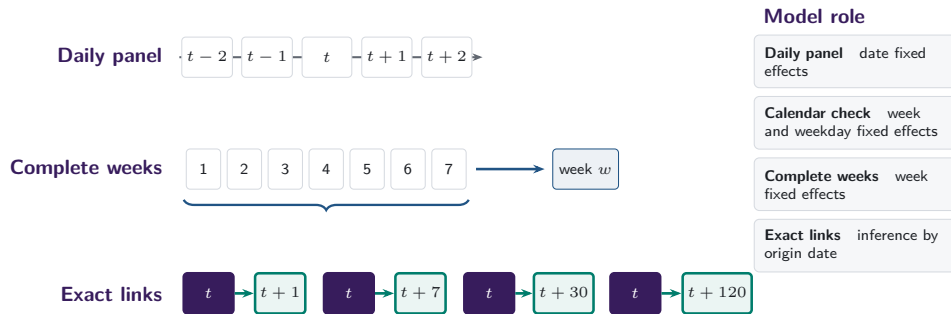
- Token-to-token execution emits two linked swaps through ETH.
- Matching ETH quantities distinguish protocol-imposed intermediation from unrelated bundled swaps.
- 129,810 mandate-era token-to-token transactions carry exactly matching ETH legs; the trace shows the largest.



same transaction, matching ETH flow

Note: Both legs of transaction 0x4dca160d...96a0ca16 (15 March 2020, block 9,674,728) report the same ETH amount to the last recorded digit.

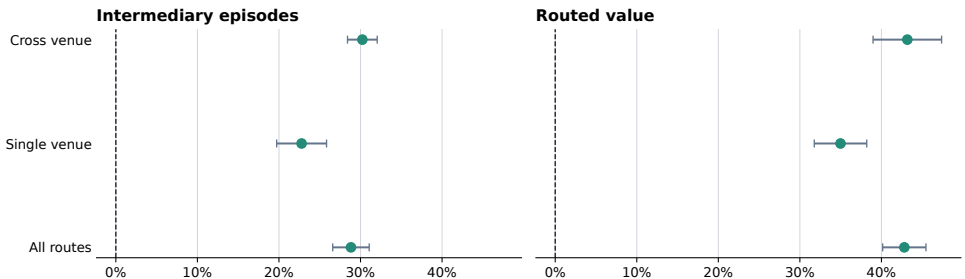
A7. Daily and weekly frequencies answer different questions



Exact horizons use the required target calendar date.

A7b. Stablecoin share rises within each venue scope

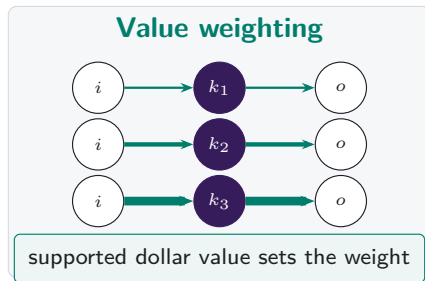
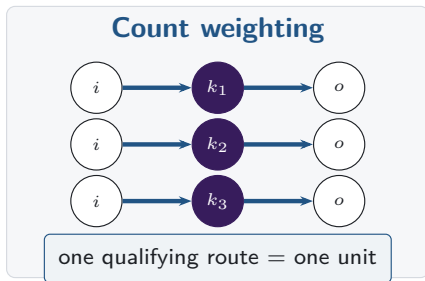
Stable share rises within realised route scopes



Points are 2024-to-2026 changes; bars are 95% HAC intervals.

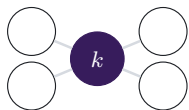
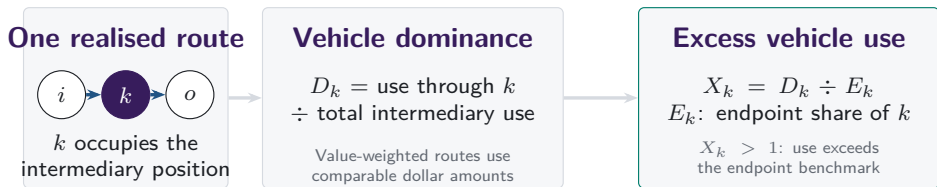
Note: Changes compare matched January-through-June dates in 2024 and 2026. Bars show calendar-HAC confidence intervals; venue scope records realised routes.

A8. Count versus value weighting

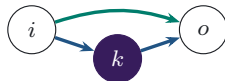


Note: Both panels contain the same routes. Line widths are illustrative, and value weighting applies only on the stated valuation-support band.

Vehicle dominance aggregates realised intermediary choices



Network position



Execution cost

Endpoint use is the benchmark; network position and same-state cost remain separate.

A9. Intermediary and route-endpoint use are separate

Intermediary share

$$I_{k,t}^N = \frac{n_{k,t}^I}{\sum_j n_{j,t}^I}$$

How often asset k sits inside a route.

Endpoint share

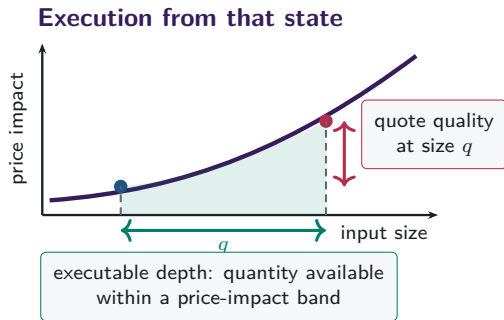
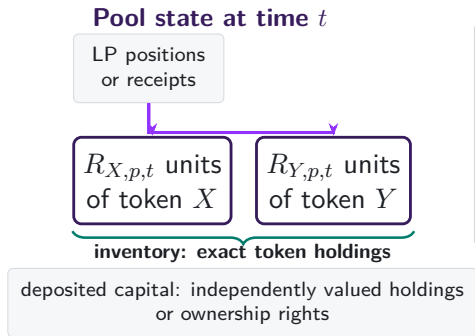
$$E_{k,t}^N = \frac{n_{k,t}^E}{\sum_j n_{j,t}^E}$$

How often k appears at either observed route endpoint.

Observed pool-route endpoints supply the benchmark.

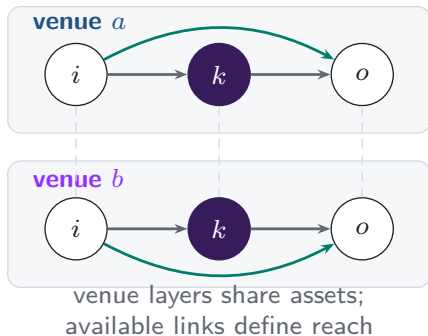
Excess use compares intermediation with observed route-endpoint use on the same support.

A10. Capital, inventory, and depth differ



A11. Additional venues expand the feasible route set

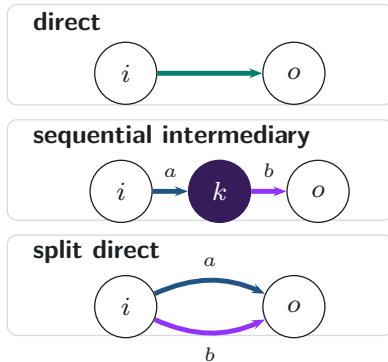
Feasible venue layers



realised
execution

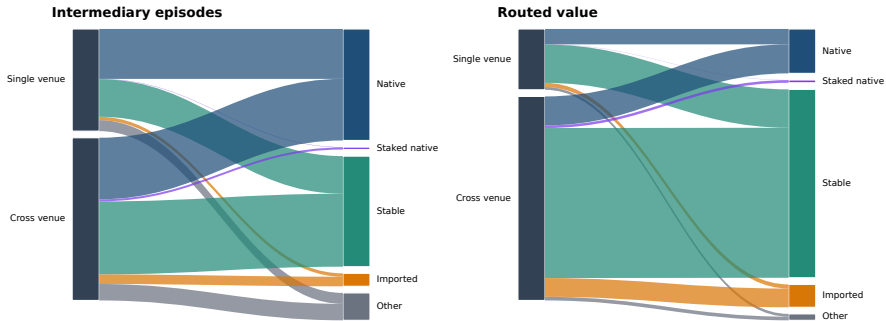


Realised route topology



A11b. Venue scope and vehicle type in 2026

Integration and vehicle composition are separate margins in 2026



Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

Note: Ribbon width is the share of 2026 intermediation jointly classified by venue scope and intermediary type.

A12. References: vehicle currencies and market structure

- Amiti, Itskhoki and Konings (2022), *Quarterly Journal of Economics*
- Dowd and Greenaway (1993), *Economic Journal*
- Flandreau and Jobst (2009), *Economic Journal*
- Gopinath and Stein (2021), *Quarterly Journal of Economics*
- Krugman (1980), *Journal of Money, Credit and Banking*
- Makarov and Schoar (2020), *Journal of Financial Economics*

A13. References: exchange design

- Adams et al. (2021), *Uniswap v3 Core*
- Adams et al. (2024), *Uniswap v4 Core*
- Angeris, Chitra, Evans and Boyd, optimal routing in constant-function markets
- Barbon and Ranaldo, decentralised exchange costs and market structure
- Lehar and Parlour (2024), *Journal of Finance*
- Millionis, Moallemi, Roughgarden and Zhang, loss versus rebalancing
- Xu, Paruch, Cousaert and Feng, AMM design and slippage

A14. WETH stays the graph hub as realised use shifts

ALL ROUTE LENGTHS; MONTHLY TRADING GRAPHS

WETH BETWEENNESS

#1

in all 8 annual graphs

0.925

in 2026 H1

WETH REALISED SHARE

76.2%

2024

42.4%

2026 H1

USDC + USDT REALISED
SHARE

14.9%

2024

37.2%

2026 H1

Binary network position changes much less than actual intermediary use.

Note: Betweenness is approximate and unweighted in annual trading graphs built from trades on the fifteenth day of each month, using 256 source nodes and a fixed seed. Realised shares use all intermediary positions on the same dates. A graph edge records an observed leg, not its depth or execution cost.

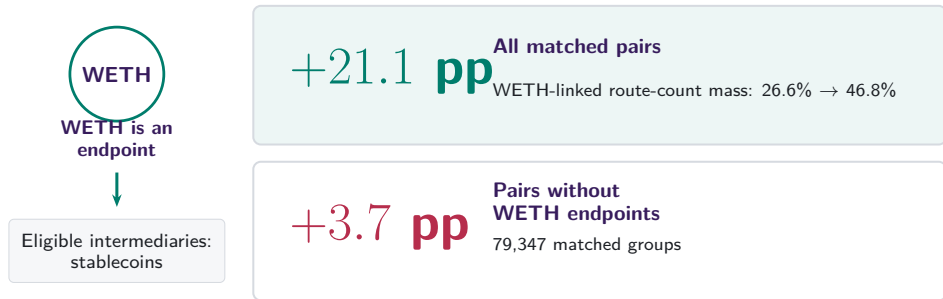
A15. Endpoint demand predicts intermediary use

	Intermediary share ($I_{a,t}$) [pp]			
	(1) Pooled	(2) Date FE	(3) + demand	(4) + currency FE
Native currency	+34.56 (23.04)	+34.55 (23.04)	+0.05 (0.56)	—
Stablecoin	+2.42* (1.32)	+2.42* (1.32)	+0.85* (0.50)	—
Endpoint-demand share ($D_{a,t}$)	— —	— —	+1.59*** (0.03)	+0.98*** (0.30)
Date fixed effects	No	Yes	Yes	Yes
Currency fixed effects	No	No	No	Yes

After date and currency effects, endpoint demand passes through almost one for one into intermediary use.

Note: 1,828,862 currency-days over 2,259 dates. Parentheses report currency/date-clustered standard errors. The outcome and demand regressor are in pp. Currency effects absorb time-invariant asset-class indicators. Asterisks *, **, and *** denote significance at 10%, 5%, and 1%, respectively.

A16. Pair composition contains a mechanical component



Changes within fixed pairs are +0.2 pp overall and +0.3 pp after removing WETH endpoints.

Note: WETH cannot be both a pair endpoint and the intermediary. Exact two-leg native-WETH-versus-stablecoin routes on common month-days; fixed-pair estimates also absorb month-day and route scope.

A17. Endpoint direction splits count and value

CONTRIBUTION TO THE STABLECOIN-SHARE INCREASE

Route count: +25.4 pp

Other pairs  +15.2 pp

One native, one stable  +9.2 pp

Two stable endpoints  +1.0 pp

Routed value: +43.9 pp

Other pairs  +20.6 pp

One native, one stable  +10.1 pp

Two stable endpoints  +13.2 pp

USDT supplies +13.7 pp of the +13.2 pp two-stable-endpoint value channel.

Note: Exact additive decomposition on 181 matched dates in 2024 and 2026. Endpoint direction describes the pair; the vehicle remains the intermediary token.

A17b. Rotation survives time and endpoint restrictions

**-0.4 pp to
+1.0 pp**
within-pair term across
all adjacent H1
comparisons

**1.1% →
9.0%**
route-count share with
nonvehicle endpoints

**0.9% →
39.1%**
supported-value share with
nonvehicle endpoints

Pair entry and exit contribute +7.2 pp by count and +33.8 pp by value.

Note: Adjacent comparisons use matched January–June dates from 2019 through 2026. The endpoint restriction removes every pair with WETH or a stablecoin at either endpoint. All changes and components are in pp; supported value applies the 20% agreement rule.

A18. Local depth remains informative alongside network reach

84.1%

of routes use the deepest supported bridge

+6.79 pp

route share per log point of weak-leg capital

+9.10 pp

route share per log point of same-day network reach

Local two-leg depth and broader network position both matter.

Note: Five vehicle currencies compared within the same set defined by pair, date, and route scope. Depth is prior-calendar deposited capital on the weaker leg. The regression absorbs set and vehicle effects and clusters by pair and date.

A19. First-use capital lies in active pools

SUPPORTED STABLECOIN, DAY -7 TO FIRST USE

92.5%

of first-use capital lies in
continuing pools

+0.16

continuing-pool growth
relative to WETH

+6.9 pp

greater probability of a
newly active pool

Note: 246 first-use events. A newly active pool has positive prior-calendar capital on day zero and none on day -7; continuing pools are positive on both dates. Difference inference clusters by pair and adoption date.

A20. V4 participation broadens after internal routing

10 PP MORE INTERNAL SAME-ASSET ROUTING

DAYS 1 TO 30

+0.086***

actions by previously
active origins

DAYS 31 TO 120

+0.153***

origins first active after
measurement

PERSISTENT VOLATILITY

+0.318***

additional later
first-active-origin
association

Note: V4 vehicle-days with 180 prior days used to classify origins. Specifications absorb vehicle and date effects, include origin-day controls, and cluster by date. Transaction origins proxy for account participation. The estimates are predictive.